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Practical No. 1

AIM: To Create and Apply Classical Cryptographic Techniques: Rail Fence Cipher and Caesar Cipher. The program should provide the user with options to select any of these ciphers to encrypt or decrypt a given message. For each cipher, the user should be able to input the necessary parameters such as keys or keywords, and the program should output the encrypted or decrypted message accordingly.

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DATE OF PERFORMANCE	27 / 06 / 26
DATE OF SUBMISSION	/ / 26
TOTAL MARKS (Performance + Viva + Lab Manual Submission)	

AIM: To Create and Apply Classical Cryptographic Techniques: Rail Fence Cipher and Caesar Cipher. The program should provide the user with options to select any of these ciphers to encrypt or decrypt a given message. For each cipher, the user should be able to input the necessary parameters such as keys or keywords, and the program should output the encrypted or decrypted message accordingly.

OBJECTIVES:

- To develop a program that implements classical cryptographic techniques, namely the Caesar Cipher and Rail Fence Cipher, for secure message encryption and decryption.
- To provide a user-friendly interface that allows users to select a cipher, enter the required keys or parameters, and perform encryption or decryption of messages efficiently.

Theory:

Cryptography is the science of protecting information by converting readable data (plaintext) into an unreadable form (ciphertext) using encryption algorithms. Only authorized users with the correct key can convert the ciphertext back into plaintext through decryption. It ensures secure communication over insecure networks by providing confidentiality, integrity, authentication, and non-repudiation. Cryptography is widely used in banking, online transactions, email security, digital signatures, and secure communication systems.

Classical ciphers are mainly classified into:

- **Substitution Ciphers** – Characters are replaced with other characters.
- **Transposition Ciphers** – Characters remain the same but their positions are changed.

In this practical, the **Caesar Cipher** (substitution cipher) and the **Rail Fence Cipher** (transposition cipher) are implemented.

1. Caesar Cipher

The Caesar Cipher is one of the oldest and simplest encryption techniques developed by **Julius Caesar**. In this method, each letter of the plaintext is shifted by a fixed number of positions in the alphabet. The number of positions shifted is called the **key**.

For example, if the key is 3:

- $A \rightarrow D$
- $B \rightarrow E$
- $C \rightarrow F$

Plaintext: **HELLO**

Ciphertext: **KHOOR**

Encryption Formula

$$C = (P + K) \bmod 26$$

Where:

- C = Ciphertext
- P = Plaintext
- K = Key

Decryption Formula

$$P = (C - K) \bmod 26$$

Advantages

- Simple and easy to implement.
- Fast encryption and decryption.
- Useful for learning basic cryptography concepts.

Disadvantages

- Very weak security.
- Easily broken using brute-force or frequency analysis.
- Only 25 possible keys.

2. Rail Fence Cipher

The Rail Fence Cipher is a **transposition cipher** that encrypts a message by writing the characters in a zigzag pattern across multiple rows (called rails). The ciphertext is obtained by reading the characters row by row.

Example (Key = 3)

Plaintext: **HELLO WORLD**

Written in zigzag:

```
H   O   L
 E L W R D
  L   O
```

Ciphertext:

HOL ELWRD LO

(The exact spacing depends on implementation.)

Working

1. Choose the number of rails (key).
2. Write the plaintext in a zigzag pattern.
3. Read each row sequentially to generate the ciphertext.
4. For decryption, reconstruct the zigzag pattern and recover the original message.

Advantages

- Easy to understand and implement.
- Does not alter the original letters.
- Demonstrates the concept of transposition encryption.

Disadvantages

- Weak against modern cryptanalysis.
- Security depends only on the number of rails.
- Not suitable for protecting sensitive information.

Code:

1) Ceaser Cipher

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
void encrypt(char text[],int shift) {
    int i;
    for (i=0;text[i]!='\0';i++){
        if(isupper(text[i])){
            text[i]= ((text[i]-'A'+ shift) % 26)+'A';
        }
        else if(islower(text[i])){
            text[i]= ((text[i]-'a'+ shift) % 26)+'a';
        }
    }
}
```

```
}
void decrypt(char text[],int shift){
    encrypt(text,26-shift);
}
int main(){
    char text[100];
    int shift, choice;
    printf("Caesar Cipher\n");
    printf("1. Encrypt\n");
    printf("2. Decrypt\n");
    printf("Enter your choice: ");
    scanf("%d",&choice);
    printf("Enter the text: ");
    scanf("%[^\\n]",text);
    printf("Enter the shift value(0-25): ");
    scanf("%d",&shift);

    shift=shift%26;
    if(choice==1){
        encrypt(text,shift);
        printf("Encrypted Text: %s\\n",text);
    }
    else if(choice==2){
        encrypt(text,shift);
        printf("Decrypted Text: %s\\n",text);
    }
    else{
        printf("Invalid Choice");
    }
    return 0;
}
```

2) Rail Fence Cipher:

```
#include <stdio.h>
#include <string.h>
void encrypt(char text[]){
    int len = strlen(text);
    char cipher[100];
    int k=0;
    //For even indice
    for(int i=0;i<len;i+=2){
        cipher[k++]=text[i];
    }
    //for odd indice
    for(int i=1;i<len;i+=2){
        cipher[k++]=text[i];
    }

    cipher[k]='\0';
    printf("Encrypted Text: %s\\n",cipher);
}
```

```
}

void decrypt(char cipher[]){
    int len =strlen(cipher);
    char plain[100];
    int fr=(len+1)/2;
    int sr=len/2;
    int i=0,j=fr;
    int k=0;

    while(i<fr || j<len){
        if(i<fr)
            plain[k++]=cipher[i++];
        if(j<len)
            plain[k++]=cipher[j++];
    }
    plain[k]='\0';
    printf("Decrypted Text: %s\n",plain);
}

int main() {
    char text[100];
    int choice;
    printf("1.Encryption \n");
    printf("2.Decryption \n");
    printf("Enter your choice: ");
    scanf("%d",&choice);
    getchar();
    printf("Enter text: ");
    fgets(text,sizeof(text), stdin);
    //Remove newline chr
    text[strcspn(text,"\n")]='\0';
    switch(choice){
        case 1:
            encrypt(text);
            break;
        case 2:
            decrypt(text);
            break;
        default:
            printf("Invalid"); return 0;
    }
}
```

Output:

The screenshot shows the Programiz C Online Compiler interface. The code in `main.c` implements a Caesar Cipher. It prompts the user to enter a text and a shift value. In the output, the user enters "Yash" and a shift of 3, resulting in the encrypted text "Bdvk".

```
main.c
20 int shift, choice;
21 printf("Caesar Cipher\n");
22 printf("1. Encrypt\n");
23 printf("2. Decrypt\n");
24
25 printf("Enter the text: ");
26 scanf("%[^\n]", text);
27 printf("Enter your choice: ");
28 scanf("%d", &choice);
29 printf("Enter the shift value(0-25): ");
30 scanf("%d", &shift);
31
32 shift=shift%26;
33 if(choice==1){
34     encrypt(text, shift);
35     printf("Encrypted Text: %s\n", text);
36 }
37 else if(choice==2){
38     decrypt(text, shift);
39     printf("Decrypted Text: %s\n", text);
40 }
41 else{
```

Output

```
Caesar Cipher
1. Encrypt
2. Decrypt
Enter the text: Yash
Enter your choice: 1
Enter the shift value(0-25): 3
Encrypted Text: Bdvk

=== Code Execution Successful ===
```

The screenshot shows the Programiz C Online Compiler interface. The code in `main.c` implements a Caesar Cipher. It prompts the user to enter a text and a choice (1 for encryption, 2 for decryption). In the output, the user enters "Yash" and chooses decryption, resulting in the decrypted text "Yash".

```
main.c
43 printf("Enter your choice: ");
44 scanf("%d", &choice);
45 getchar();
46 printf("Enter text: ");
47 fgets(text, sizeof(text), stdin);
48 //Remove newline chr
49 text[strcspn(text, "\n")] = '\0';
50 switch(choice){
51     case 1:
52         encrypt(text);
53         break;
54     case 2:
55         decrypt(text);
56         break;
57     default:
58         printf("Invalid");
59 }
60 return 0;
61 }
62
63
64
```

Output

```
1.Encryption
2.Decryption
Enter your choice: 2
Enter text: Yash
Decrypted Text: Yash

=== Code Execution Successful ===
```

Conclusion:

This practical helped in understanding the fundamental concepts of cryptography and information security. It provided knowledge about encryption, decryption, security services, and different types of cryptographic algorithms. The virtual laboratory also offered practical exposure to cryptographic techniques, making it easier to understand their real-world applications in securing digital communication.

DISCUSSION QUESTIONS

1. What is the Caesar Cipher?
2. What is the main difference between substitution and transposition ciphers?
3. Why is the Caesar Cipher considered insecure?
4. What type of cryptographic technique is the Rail Fence Cipher?
5. What is the Rail Fence Cipher?

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EVALUATION:

Submission and Viva Marks (3)	
Faculty Signature & Date	